Day 058 | Unit 04 | Teacher plan

Numeric and Algebraic Expressions

Identify Equivalent Expressions

Standards: 6.EE.A.4

Learning target: I can justify whether expressions are equivalent.

enVision Topic 3: Numeric and Algebraic Expressions | Spectrum companion Chapter 5

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Use operation properties to rewrite an expression. Matching values at one input do not prove equivalence for all inputs. A single counterexample is enough to show two expressions are not equivalent.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 8

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01 / Learn

Use operation properties to rewrite an expression. Matching values at one input do not prove equivalence for all inputs. A single counterexample is enough to show two expressions are not equivalent.

Worked example 1

Are 6(x + 7) and 6x + 42 equivalent for every x? Use a property or a counterexample.

1. Use operation properties to rewrite an expression.

2. Matching values at one input do not prove equivalence for all inputs.

3. 6(x + 7) = 6x + 42.

4. This matches.

5. State the result in the form requested: Yes; distribution gives 6x + 42.

Answer: Yes; distribution gives 6x + 42

Worked example 2

Select all expressions equivalent to 6(x + 3).
A. 9x
B. 6x + 18
C. 6x + 3
D. 18 + 6x

1. Use operation properties to rewrite an expression.

2. Distribute 6 to both terms to get 6x + 18.

3. Reversing the addends gives the same value.

4. The other expressions fail, for example at x = 0.

Answer: B, D - 6x + 18; 18 + 6x

Day 058 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Are 7(x + 2) and 7x + 2 equivalent for every x? Use a property or a counterexample.

02 / Constructed response

Are 2(x + 5) and 2x + 10 equivalent for every x? Use a property or a counterexample.

03 / Constructed response

Are 8(x + 7) and 8x + 7 equivalent for every x? Use a property or a counterexample.

04 / Constructed response

Are 8(x + 2) and 8x + 16 equivalent for every x? Use a property or a counterexample.

05 / Constructed response

Are 3(x + 2) and 3x + 2 equivalent for every x? Use a property or a counterexample.

06 / Constructed response

Are 2(x + 7) and 2x + 14 equivalent for every x? Use a property or a counterexample.

07 / Constructed response

Are 6(x + 6) and 6x + 6 equivalent for every x? Use a property or a counterexample.

08 / Multiple select

Select all expressions equivalent to 7(x + 4).
A. 11x
B. 7x + 28
C. 7x + 4
D. 28 + 7x

09 / Multiple select

Select all expressions equivalent to 4(x + 6).
A. 10x
B. 4x + 6
C. 4x + 24
D. 24 + 4x

10 / Constructed response

Are 3(x + 6) and 3x + 18 equivalent for every x? Use a property or a counterexample.

Day 058 | Unit 04 | Answer key

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1. No; at x = 0, values are 14 and 2

7(x + 2) = 7x + 14. The constants 14 and 2 differ.

6.EE.A.4 | Constructed response

2. Yes; distribution gives 2x + 10

2(x + 5) = 2x + 10. This matches.

6.EE.A.4 | Constructed response

3. No; at x = 0, values are 56 and 7

8(x + 7) = 8x + 56. The constants 56 and 7 differ.

6.EE.A.4 | Constructed response

4. Yes; distribution gives 8x + 16

8(x + 2) = 8x + 16. This matches.

6.EE.A.4 | Constructed response

5. No; at x = 0, values are 6 and 2

3(x + 2) = 3x + 6. The constants 6 and 2 differ.

6.EE.A.4 | Constructed response

6. Yes; distribution gives 2x + 14

2(x + 7) = 2x + 14. This matches.

6.EE.A.4 | Constructed response

7. No; at x = 0, values are 36 and 6

6(x + 6) = 6x + 36. The constants 36 and 6 differ.

6.EE.A.4 | Constructed response

8. B, D - 7x + 28; 28 + 7x

Distribute 7 to both terms to get 7x + 28. Reversing the addends gives the same value. The other expressions fail, for example at x = 0.

6.EE.A.4 | Multiple select

9. C, D - 4x + 24; 24 + 4x

Distribute 4 to both terms to get 4x + 24. Reversing the addends gives the same value. The other expressions fail, for example at x = 0.

6.EE.A.4 | Multiple select

10. Yes; distribution gives 3x + 18

3(x + 6) = 3x + 18. This matches.

6.EE.A.4 | Constructed response